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Initial Post

by [Ali Alhammad](#) - Tuesday, 26 August 2025, 8:17 PM

The communication aspect of agents in the field of intelligent system research has been key in attaining cooperation and coordination. To address this, agents have embraced Agent Communication Languages (ACL) of which one is the use of Knowledge Query and Manipulation Language (KQML). In comparison with conventional message passing, ACLs are more concerned with the transfer of knowledge and intention, as such multi-agent systems (MAS) will be able to behave in open, dynamic, heterogeneous environments (Ma et al., 2024). The general greatest strength of ACLs lies in the fact that they enable standardising communication between different agents. Such compatibility is needed in a distributed application such as e-commerce, supply-chain management, and knowledge-sharing systems (Allouch et al., 2021). Other than that, ACLs can render MAS scalable in that addition of new agents to the system will not disrupt the activities present in the existing systems. This makes them particularly helpful in problem-solving situations which may be complex (Cao et al., 2024). Although conceptually appealing, there are some flaws to ACLs. They need sophisticated reasoning capability, which further complicates the design of systems. The other challenge is the processing cost that results from the parsing and interpretation of performatives (Kuhail et al., 2023). ACLs are slower and less efficient than low-level approaches and have limited practical utility in time-dependent areas such as robotics or real-time analytics. The trade-offs are abundant since they are compared to method invocation in Python or Java. The invocation of methods is direct, lightweight, efficient, and thus suitable in tightly coupled software environments. However, it does not allow the abstraction and autonomy provided by ACLs. When throughput is less important than flexibility, and when systems are loosely connected in that agents may not be familiar with internal mechanisms of other agents, CLSs can be more flexible at the cost of speed, and then CLSs use CLSs (ACLs) (Liu et al., 2025). Languages such as KQML are a more ambitious attempt at successful intelligent-agent communication. ACLs are still useful in research, and though the industry usually prefers simpler options for efficiency reasons, they continue to shape current thinking about interoperability and intelligent systems design.

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Peer Response

by [Abdulla Husain Salem Hadna Almessabi](#) - Monday, 1 September 2025, 4:48 AM

Hi Ali,

Your post gives a concise description of the trade-offs between ACLs such as KQML and the traditional approach to method invocation. One of the greatest strengths of the ACLs is that they allow the standardisation of communication between a heterogeneous and distributed system (Moore, 2000). This is crucial in such areas as e-commerce and supply-chain management, where various agents, usually designed with different architectures, still need to work together (Labrou and Finin, 1998). The scalability feature you discussed matters, too, as multi-agent systems can typically be expected to dynamically scale by introducing new agents without re-architecting the entire communication layer.

Your criticism of ACL limitations was also quite good in my opinion. Their parsing and reasoning overhead can often render them impractical for time-sensitive applications like robotics. Finin et al. (1994) stated that, though conceptually powerful, the cost of ACLs is less compelling in real-time scenarios. This tension is well brought out in your comparison of Python and Java's approach to method invocation: direct calls are fast but do not provide the open system freedom and abstraction.

Another possible approach might be the incorporation of the ACL principles in contemporary middleware. As an example, standards like the JADE or FIPA standards are trying to fill this gap between ACL-style semantics and more lightweight message-passing infrastructures (Bouzoubaa and Moulin, 1998). Such a hybrid solution can contribute to reducing the efficiency issues without loss of standardised agent interaction advantages (Prymek and Horak, 2008).

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Peer Response

by [Mansour Saeed Mubarak Hamdan Al Hamdani](#) - Wednesday, 3 September 2025, 5:54 PM



Your discussion on Agent Communication Languages (ACLs), particularly KQML, effectively highlights their role in fostering cooperation and coordination in multi-agent systems (MAS). I agree that ACLs' focus on knowledge and intention transfer, as noted by Ma et al. (2024), enables them to thrive in open, heterogeneous environments, making them ideal for applications like e-commerce and supply-chain management (Allouch et al., 2021). The scalability you mention, supported by Cao et al. (2024), is a significant advantage, allowing seamless integration of new agents.

Your analysis of ACLs' drawbacks—requiring sophisticated reasoning and incurring processing costs (Kuhail et al., 2023)—is insightful. The inefficiency compared to method invocation in Python or Java, which excels in tightly coupled systems, underscores a clear trade-off (Liu et al., 2025). Your point about ACLs' limited use in time-sensitive fields like robotics, favoring simpler industry solutions, aligns with current trends.

To build on this, it could be worth exploring how advancements in AI might reduce ACLs' computational overhead or enhance their real-time applicability. Your balanced view on flexibility versus speed is compelling—great work, and I'm eager to hear more on future ACL innovations.

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Peer Response

by [Fahad Abdallah](#) - Saturday, 6 September 2025, 7:31 AM

Hello Ali, your post gives a clear and organised summary of how Agent Communication Languages (ACLs), particularly KQML, can facilitate communication in multi-agent systems (MAS). I concur with your point that ACLs go beyond mere syntactic message passing to concentrating on the semantics of interaction so that agents can exchange data, intentions, and commitments. The given view is backed by the recent research, which highlights that the agents created using large language models rely highly on the communication frameworks to organise the knowledge and unlock the collective intelligence in the complex systems (Li et al., 2023).

I also liked how you focused on heterogeneous environments. ACLs provide a platform of interoperability, an important element in fields like supply-chain management and distributed applications, by facilitating standardisation across a wide range of platforms and programming languages. The latter echoes the discoveries in multi-agent reinforcement learning, where scalability and coordination issues of the dynamical environment are addressed by proper communication protocols (Canese et al., 2021).

Your discussion of ACL limitations was well-balanced. Their complexity of computation can make them inapplicable to real-time (or high-performance) systems due to the requirement of complex reasoning mechanisms. The problem is also well understood in the existing literature on multi-agent reinforcement learning in uncertain and noisy settings, where communication efficiency is

essential to the system's reliability (Tung et al., 2021).

An opportunity that can be tapped is in hybrid models. Recent advances in multi-agent systems integrate logic-based communication frameworks with machine learning approaches (Li et al., 2023; Canese et al., 2021). For example, logic-based representations (Calegari et al., 2021) with learning based communication can allow the agent to balance efficiency and interpretability. This implies the possibility of ACLs becoming more useful as a tool for real-life situations, especially where autonomy and versatility are needed.

Your post effectively highlights the potential of the ACLs as well as their challenges, and linking them to the research enhances your analysis further.

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Peer Response

by [Ali Alzahmi](#) - Saturday, 6 September 2025, 7:47 AM

Ali, in your post, it is evident how the Agent Communication Languages (ACLs) and specifically KQML facilitate agent-agent communication by emphasising knowledge and intention transfer, not mere message transmission. I agree that the difference makes multi-agent systems (MAS) effective in open and heterogeneous environments, where flexibility and interoperability are needed. Recent surveys on multi-agent systems based on LLMs confirm that structured communication is among the most significant aspects of letting the agents cooperate and adapt to the changing environment (Li et al., 2024).

It is also significant how you highlight standardisation as a strength of ACLs. As you described, distributed applications like e-commerce, supply chain management, and knowledge-sharing must be compatible with various systems. ACLs improve scalability and interoperability because they deliver a common communication layer. This aligns with the larger research patterns that emphasise that agents engineered by other organisations can work together more effectively if communication protocols become standardised (Liu et al., 2023).

Meanwhile, your trade-offs were correctly recognised. ACLs need high reasoning, which makes them complex and slow to compute. They are also unsuitable in real-time or performance-sensitive systems, such as robotics, because parsing and interpreting performatives have overhead. These disadvantages make lightweight methods like Python or Java method invocation a favorite in industries because they are fast and efficient. However, as you observed, these methods are too finite to provide the abstraction and independence required in loosely coupled systems where agents must decode one another.

Lastly, it is clear why ACLs continue to dominate the research world, which is why they are so significant. Simpler variants prevail in the commercial arena; however, ACLs still directly impact the conceptualisation of interoperability and intelligent system design in academia and professional environments (Getchell et al., 2022).

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